

3. DETERMINATION OF WAVELENGTH OF LASER SOURCE- DIFFRACTION GRATING

Aim:

To determine the wavelength of given laser light by using diffraction grating.

Apparatus:

A diode laser, transmission grating, thread, wooden scale, an optical bench, screen.

Principle:- When laser beam is transmitted through a series of narrow parallel slits that are spaced together, each slit produces a diffraction pattern. The diffraction pattern produced by one slit overlap with the patterns produced by the other slit. Light coming from the multiple slits produces a much brighter pattern.

Formula:

The wave length of the laser light used is

$$\lambda = \frac{ay}{mD} \text{ \AA}$$

Where, a = width of the slit,

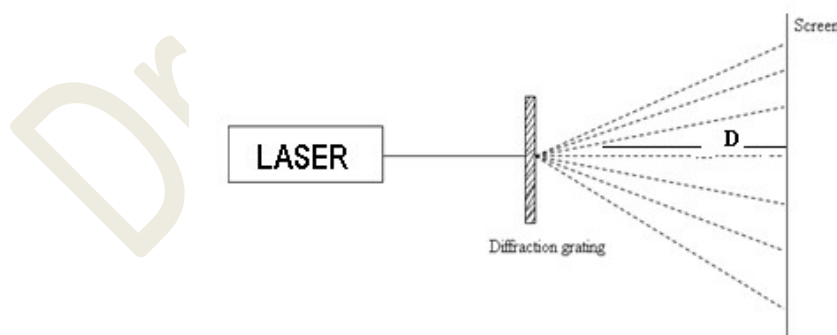
D = the distance from grating to screen,

m = order diffraction,

y = distance on the screen from the center spot to the next spots.

Experimental Procedure:

A laser source is arranged on an optical bench. The screen is fixed nearly 1m away from the laser. The diffraction grating is mounted in the optic bench in such a way that it should lie in between the screen and the laser as shown in fig. The diffraction grating is slightly adjusted so that the rulings 2500 lines/inch is in line with the laser beam and the grating is adjusted so as to get a clear diffraction pattern on the screen. By walking close to the screen and carefully observing the diffraction pattern, the distance between any two adjacent slits is measured.



The distance between the slit (grating) and the screen is measured.

The measured values are tabulated, the wave length of laser beam is determined.

Using Grating:

Observations:

Table 1:

D =

a =

S.No	order diffraction (m)	Distance of m th order from zero th order		Average y (cm)	$\lambda = \frac{ay}{mD} \text{ \AA}$
		Left (cm)	Right (cm)		
1					
2					
3					
4					
5					

Average $\lambda =$ _____ $\text{\AA}$

Table 1: D =

a =

S.No	order diffraction (m)	Distance of m th order from zero th order		Average y (cm)	$\lambda = \frac{ay}{mD} \text{ \AA}$
		Left (cm)	Right (cm)		
1					
2					
3					

Average $\lambda =$ _____ $\text{\AA}$

Calculations:

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Precautions:

1. We should not see the laser beam directly with naked eyes
2. Be careful while observing the diffraction pattern.
3. Distance should be measured without any parallax error.

Result: The wavelength of laser light $\lambda =$ _____ cm
= _____ $\text{\AA}$

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